

PROJECTS GROUP 4: INDOOR SPORTS CLUB SYSTEM
ANISH AND GIDEON

Narrative

We plan to build a system that manages our sports club operations more efficiently. Instead of paper records, we want members to register, pay fees, and book badminton, squash, or tennis courts online or at reception. We will give priority to premium members and also manage blocked slots for tournaments or maintenance.

We want coaches to manage their schedules in the system, and all lessons and payments to be tracked. We also expect the system to handle cancellations, reschedules, and generate weekly reports on court usage, income, member activity, and coach schedules. This will reduce conflicts, prevent missed payments, and improve service for our members.

INPUT DOCUMENTS

Members:

MemberID	
MemberName	
Phone	
Email	
Subscription	
Subscription	

Courts:

CourtNumber	
CourtType	
CourtCapacity	
CourtAvailability	

Coaches:

CoachName	
CoachID	
Phone	
Email	
CoachSport	
CoachAvailability	
CoachRate	

Booking:

BookingNumber	
BookingType (coach/court)	
MemberBooking (member/occasional)	
BookedDateAndTime	
TimeOfBooking	

Payments:

PaymentID	
PremiumRateMonthly	
PremiumRateAnnually	
PaymentType PaymentDateAndTime	
CancellationFee (24 hour limit)	

Tournaments:

CourtsBooked	
TimeBooked	
TournamentDateAndTime	

Routine Reports:

NumberOfMembers	
MembershipType	
NoOfPremiumsMonthly	
NoOfPremiumsAnnually	
courtUsage	
NumOfCoachingLessons	

Some ideas

We can establish a revenue breakdown report for just money alone : Membership fees, booking fees, coaching fees.

We can also create a coach schedule report. How many lessons and hours the coach used. It can integrate it under NumOfCoachingLessons or we can make a new table either is fine

Would it be a good idea to give each member an activity report?

- This we can keep in mind for the future if our club does well and we have enough resources to focus on individual users.

SCREEN REPORTS

1. **Membership Screen**

Options: Register New Member, Renew Membership, View Membership Status.

Display: Active/expired members list.

2. **Court Booking Screen**

Options: Search Courts by Date/Time, Book Court, Cancel/Reschedule Booking.

Display: Court availability calendar, booking confirmation.

3. **Coach Scheduling Screen:**

Options: Assign Coach Session, Cancel Session, View Coach Calendar.

Display: Coach availability, member assignment.

4. **Payments Screen**

Options: Record Payment, View Payment History, Print Receipt.

Display: Pending payments, receipts generated.

5. **Tournament Management Screen**

Options: Reserve Courts for Tournament, View Reserved Slots.

Display: Tournament calendar blocking normal bookings.

6. **Reports Dashboard**

Options: Generate Court Utilization, Revenue, Coach Schedules, Member Activity Reports.

Display: Download/export reports in PDF or Excel.

EXAMPLE / SAMPLE OF WHAT SCREEN WOULD LOOK LIKE:

Options Available:

- Search Courts by Date/Time
- Book Court (Options for tournaments or regular sessions)
- Cancel/Reschedule Booking
- Calendar view of available slots
- Dropdown for sport type (Badminton, Tennis, Squash)
- Court availability in real-time

Workflow (Step-by-Step):

1. Members log into the system.
2. Selects "Book Court."
3. Choose TENNIS from the dropdown.
4. Calendar shows available slots , member clicks Aug 26, 5:00–6:00 PM.
5. Clicks Confirm Booking.
6. The system displays a message: "Booking Confirmed – Court C2 (Tennis), Aug 26, 5–6 PM" and generates a receipt.

Sample what the Screen Shows:

The mockup shows a user interface for booking a tennis court. At the top, there are two dropdown menus: "[Sport : (Tennis) v]" and "[Date : (Aug 26) v]". Below these is the heading "Available Time Slots :". A table follows with two rows: the first row shows time slots "9-10 AM", "10-11 AM", "11-12 AM", and "..." with corresponding status indicators "Free", "Booked", "Free", and "..."; the second row shows the same time slots with status indicators "Free", "Booked", "Free", and "...". Below the table are three buttons: "[Book Court]", "[Cancel Booking]", and "[Reschedule]". A horizontal dashed line separates the booking options from the confirmation message. The confirmation message reads: "Booking Confirmation", "Court : Tennis Court 2", "Date : Aug 26, 2025", "Time : 5:00 - 6:00 PM", and "Status : CONFIRMED".

ASSUMPTIONS

Members can hold either monthly or annual memberships only.

Premium members receive priority booking rights.

Courts can only be booked in hourly slots (no half-hours).

A booking cannot overlap with tournament reservations or maintenance.

Coaches are assigned to one session at a time (no double-booking).

Payments must be completed before a booking or coaching session is confirmed.

Cancellation and rescheduling are allowed up to 24 hours before the booking time.

The system automatically blocks bookings for expired memberships.

Daily and weekly reports reset after being generated (archived separately).

Edge Case: If a member tries to book during a blocked tournament slot -> system rejects with message: "Court unavailable due to tournament."

Edge Case: If payment fails → system does not confirm the booking.

In case a person books a court they have the option to book the coach assigned to the court and possibly also assigned to the court. If the coach isn't booked by whoever needs the court but another needs the coach we assume there is more space or another room available to use with the right equipment